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Section - 4 Group - 1

Michelson's Interferometer

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Aim of the Experiment

To determine the wavelength of the given source using the phenomenon of interference based on division of amplitude.

Apparatus Required

(i) Michelson's interferometer set up.

(ii) Light source (He-Ne Laser 1.0 mW, 220 V AC)

(iii) Lens mounted, f=20 mm with holder

(iv) Optical bench and Screen

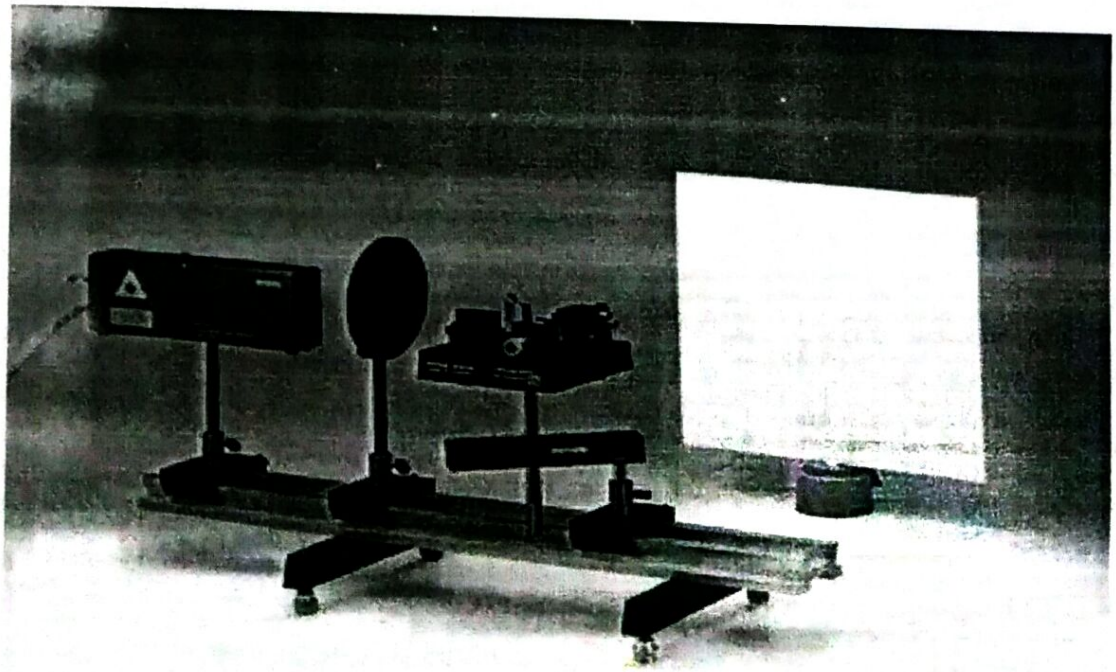

Fig-1: Experimental Set-up

Theory

When two coherent sources of light interfere with each other we get an interference pattern. As a result of this a new wave is generated with modification in the distribution of intensity through superposition. Two sources are said to be coherent if there exists a constant phase relationship in between the two sources. The interference pattern is produced in terms of point of maximum intensity (maxima) and minimum intensity (minima). In case of Michelson's interferometer interference takes place by division of amplitude. The complete experimental set up for Michelson's interferometer is shown in Fig-1.

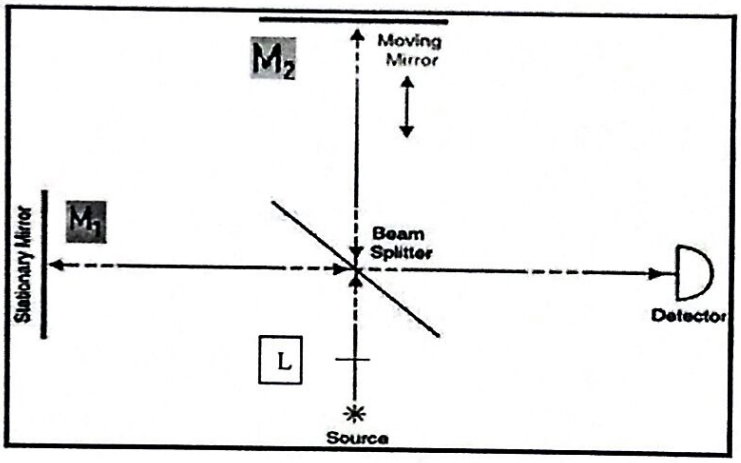

Fig-2: Schematic diagram

The schematic diagram is illustrated in Fig-2. Basically the set up consists of lens L, a half silvered glass plate, two mirrors M_1 and M_2 with the detector. Light from a monochromatic laser source after passing through the lens L falls on the half silvered glass plate called as beam splitter, which is inclined at an angle 45^0 to the path of light. Light falling on it is partly reflected towards mirror M_1 and partly transmitted towards M_2.The beam going towards M_1 is reflected back normally and is transmitted through the glass plate to fall on the detector. Similarly the light got transmitted through the glass plate is reflected

back by the mirror M_2 and again strikes on glass plate to get reflected and reach the detector. The mirrors M_1 and M_2 are adjusted to get circular fringes as shown in Fig-4. The mirror M_2 is kept fixed and the mirror M_1 is moved by the micrometer screw attached to it. The number of circular fringes crossed the field of view is counted and the corresponding distance moved on linear scale is noted.

If λ= Wavelength of the laser source of light.

x = the distance through which the micrometer screw moved

n = No of fringes that crossed the centre of the field of view or the no of fringes that collapse in the centre. Then $\lambda = \frac{2x}{n}$

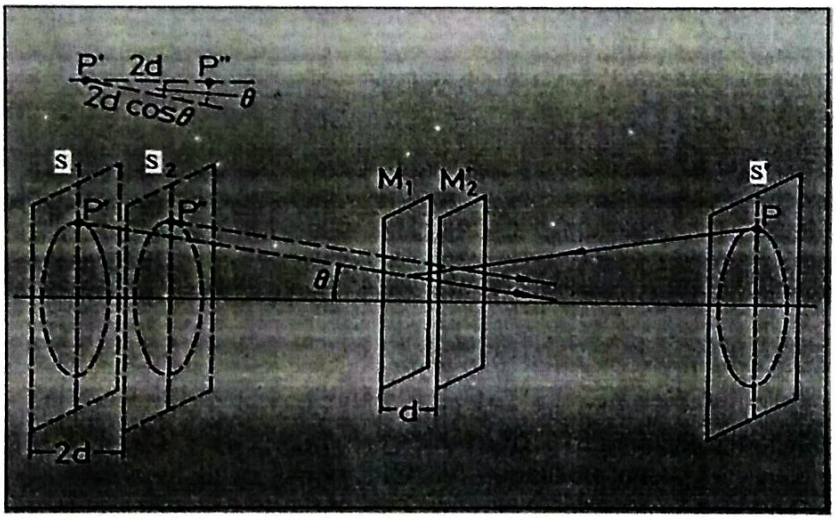

Fig-3: Schematic diagram

As shown in the Fig-3 M_2' is the image of the mirror M_2 on the glass plate and S_1, S_2 are the images of the S' on the same glass plate. We have replaced the real source S by its image S'. Here P, P' and P" are three points on S', S_1 and S_2 respectively. If the separation between M_1 and M_2' is d then the separation between S_1 and S_2 is 2d. These two images of the sources serve as two virtual coherent sources. The rays coming from these sources $S_1$and S_2 and making an

angle θ with the central line interfere with each other with a path difference $2d\cos\theta$. Depending on the value of the path difference a particular fringe will be dark or bright.

For $2d\cos\theta = n\lambda$, we get a maxima and for $2d\cos\theta = (2n+1)\lambda/2$, we get a minima where n is the order of the bright or dark fringes and λ is the wave length of the light used.

By moving the mirror M_2 through a distance x and counting the number of bright fringes m that merge at the centre of the circular fringe pattern, the value of the wave length can be calculated .

We have $2d\cos\theta = n\lambda$ ---------- (1)

After moving the mirror through x,

$2(d+x)\cos\theta = (n+m)\lambda$ ---------- (2)

Subtracting eqn (1) from eqn (2) we got,

$2x\cos\theta = m\lambda$ ---------- (3)

For all the rays passing through the centre, $\theta = 0$.

So eqn (3) becomes as, $2x = m\lambda$

Or $\lambda = 2x/m$

In the present set up the actual distance moved by the mirror is 1/10th of the distance moved by the micrometer screw head. So $\lambda = 2x/10m$. From this the value of λ can be calculated.

Fig-4: Interference Pattern obtained from Michelson's Interferometer

Procedure

(i) Mount the laser source and switch it on for some time so that the output light gets stabilized. Adjust the lens on the optical bench and focus the laser light on the half-silvered mirror.

(ii) The resulting two beams of light are reflected by the two mirrors and impinge on the screen. With the help of the adjusting screws provided move the positions of the two images and make them interfere to give circular fringes.

(iii) Now turn the micrometer screw to any initial position at which the centre of the circles is dark and note down the reading. Again turn the micrometer in the same direction and the no of fringes crossed the centre of the field of view is counted.

(iv) The distance travelled on the micrometer scale is noted and from this the value of d is calculated by dividing 10 to the fringe width.

(v) Finally calculate the wavelength of the source using the given formula.

Observations

Table-1: Measurement of 'x' for 50 fringes:

Sl. No.	I.C.S.R I	No of Complete rotation(n)	F.C.S.R F	Pitch (mm)	L.C. (mm)	P.S.R A=Pitch*n (mm)	Circular scale reading B=(I-F)*L.C. (mm)	Total Path diff. A+B=F (mm)	x=F/10 (mm)
1	12	0	29	0.5	0.01	0	0.17	0.17	0.017
2	29	0	46	0.5	0.01	0	0.17	0.17	0.017
3	46	0	13	0.5	0.01	0	0.17	0.17	0.017

4	13	0	29	0.5	0.01	0	0.16	0.16	0.016
5	29	0	45	0.5	0.01	0	0.16	0.16	0.016
6	45	0	12	0.5	0.01	0	0.17	0.17	0.017
7	12	0	28	0.5	0.01	0	0.16	0.16	0.016
8	28	0	45	0.5	0.01	0	0.17	0.17	0.017
9	45	0	11	0.5	0.01	0	0.16	0.16	0.016
10	11	0	28	0.5	0.01	0	0.17	0.17	0.017

Where I.C.S.R = Initial circular scale reading

F.C.S.R. = Final circular scale reading

Results and calculations

Calculation of Least count

Here 1 M.S.D. = 1mm = 2 rotations of circular scale

1 rotation of circular scale = 0.5mm = 50 divisions

So 1 division of circular scale = 0.01mm = L.C.

The wave length of the laser light is calculated using the formula,

$$\lambda = \frac{2x}{n}$$

Conclusions

Conclusion should incorporate

a. Interesting results you found in the experiment.

b. Lacunas /difficulties that you encountered.

c. Practical applications/implications of the experiment.

d. Any other issues.

Precautions

(ii) One should not look directly to LASER light because it may damage the eye.

(iii) One should carefully handle the interferorometer because slight disturbance may disturb the fringe obtained.

(iv) After finishing the experiment the interferorometer should covered properly.

References

1. Fundamentals of optics by F. Jenkins and H. White.
2. Optics by A. Ghatak.
3. Introduction to Optics by E. Hecht
4. Feynman Lectures.

Considerations

(iv) What is interference of light and what are the different ways in which light interferes?

(v) What is an interferometer and what is the history of Michelson's interferometer?

(vi) Name different types of interferometers and their applications?

Calculation

For $x_1 = 0.017$ mm

$$\lambda_1 = \frac{2 \times x_1}{n} = \left(\frac{2 \times 0.017}{50}\right) \text{mm}$$

$$= 680 \text{ nm}$$

For $x_2 = 0.016$ mm

$$\lambda_2 = \frac{2 x_2}{n} = \frac{2 \times 0.016}{50}$$

$$= 640 \text{ nm}$$

$$\bar{\lambda} = \frac{680 + 680 + 680 + 640 + 640 + 680 + 640 + 680 + 640 + 680}{10}$$

$$= 664 \text{ nm}$$

$$d_1 = \bar{\lambda} - \lambda_1$$

$$= (664 - 680) \text{ nm} = -16 \text{ nm}$$

$$d_2 = \bar{\lambda} - \lambda_2$$

$$= (664 - 640) \text{ nm} = 24 \text{ nm}$$

$$S^2 = \frac{\sum_i d_i^2}{n} = \frac{(-16)^2 + (-16)^2 + (-16)^2 + 24^2 + 24^2 + (-16)^2 + 24^2 + (-16)^2 + 24^2 + (-16)^2}{10}$$

$$= 384$$

$$\sigma = \sqrt{\frac{S^2}{n-1}} = \sqrt{\frac{384}{9}}$$

$$= 6.53 \text{ nm}$$

$$\lambda = \bar{\lambda} + \sigma = (664 \pm 6.53) \text{ nm}$$

① Conclusion:

① The wavelength of light wave of obtained is (664 ± 6.53) nm.

② The concept of Michelson's Interferometer can be used in LIGO to detect gravitational waves.

③ There is a certain error margin arising due to the defects in mirror and slight variation in the lengths of paths.